A Few Notes on Cayley's Theorem

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1 Introduction

Group Theory is the study of symmetry. Cayley's Theorem is a fundamental theorem in Group Theory, and the topic of these notes.

Before diving into Cayley's Theorem, a couple of notes:

- The Symmetric Group $Sym(G)$ or sometimes S_n, where $n = |G|$ (G is finite), is the set of all bijections from G to itself with function composition as the group operation. That is, $Sym(G) = S_n = S_{|G|} = \{f : G \to G \mid f \text{ is an bijection}\}$.
- We use the symbol $\simeq$ (or sometimes $\cong$) to mean that the groups G and H are *isomorphic*. That is, $G \simeq H \implies \exists f \mid f : G \to H$ where f is a bijection and a homomorphism. See Equation 1.
- To show that f is one-to-one, show that $f(x) = f(y) \implies x = y$.
- To show that f is onto, pick an arbitrary $h \in H$ and show that $\exists g \in G \mid f(g) = h$.

Recall also that if we have two groups $(G, *)$ and $(H, \cdot)$ we say that $(G, *)$ is *isomorphic* to $(H, \cdot)$ if there exists a bijection $f : G \to H$ which satisfies the *homomorphism* property:

$$f(x * y) = f(x) \cdot f(y) \quad \forall x, y \in G \tag{1}$$

That is, f is a bijection (one-to-one and onto) and f is also a homomorphism.

Any bijective function f which satisfies Equation 1 is called a *group* isomorphism from G to H. The basic idea of $(G, *)$ being isomorphic to $(H, \cdot)$ is that $(G, *)$ and $(H, \cdot)$ are "algebraically equivalent". That is, there is a one-to-one correspondence between elements of G and elements of H where the outcomes of operations on elements of G are matched with the outcomes of the corresponding operations on the corresponding elements of H.

2 Cayley's Theorem

Theorem 2.1. Cayley's Theorem: If G is a group then there exists a subgroup H of $Sym(G)$ such that G is isomorphic to H.

Proof: Suppose that G is a group. Then to prove Cayley's Theorem we need to find a subgroup H of $Sym(G)$ and a bijective homomorphism $f : G \to H$. My roadmap for the proof looks like

1. Define $\phi_a : G \to G$ for each $a \in G$ and show that ϕ_a is a bijection
2. Define $H = \{\phi_a \mid a \in G\}$ and show that H is a subgroup of $Sym(G)$
3. Define $f : G \to H$ and show that f is both a bijection and a homomorphism

BTW, a nice thing about the proof of Cayley's theorem is that it is a *constructive* proof: the statement of the theorem is that a certain group H exists. In the course of the proof of the theorem one can actually show not only that such an H exists but also how to actually find it. We'll see an example of this below (Section 3.2).

2.1 Define $\phi_a : G \to G$ for each $a \in G$ and show that ϕ_a is a bijection

To start, for each fixed element $a \in G$ define $\phi_a : G \to G$ by the map $x \mapsto ax$. That is

$$\phi_a(x) = ax \quad \forall x \in G \tag{2}$$

Luckily it turns out that each ϕ_a is a bijection. To see this we need to show that ϕ_a is one-to-one and onto. First, consider that ϕ_a is one-to-one since

$$\begin{aligned}
\phi_a(x) &= \phi_a(y) && \text{\# to show } \phi_a \text{ is 1-to-1 show } \phi_a(x) = \phi_a(y) \Rightarrow x = y \\
&\Rightarrow ax = ay && \text{\# definition of } \phi_a(x) \text{ (Equation 2)} \\
&\Rightarrow a^{-1}(ax) = a^{-1}(ay) && \text{\# multiply by } a^{-1}; a \in G \text{ \& } G \text{ a group} \Rightarrow a^{-1} \in G \\
&\Rightarrow (a^{-1}a)x = (a^{-1}a)y && \text{\# multiplication is associative} \\
&\Rightarrow x = y && \text{\# } a^{-1}a = 1
\end{aligned} \tag{3}$$

So ϕ_a is one-to-one.

Aside on cancellation laws: Note that in (3) we used the fact that $a \in G$ and that G is a group so $a^{-1} \in G$. Here we have $a^{-1}a = 1$, which essentially gives us a *cancellation law*[1]; in (3) this allows us to "cancel" the a on both sides. Now, what if we don't have access to multiplicative inverses? We might be faced with this situation if we have a ring, where we don't in general have multiplicative inverses[2]. So if we don't have multiplicative inverses how do we go about showing that something is one-to-one?

One approach is to factor out a and note that by assumption, $a \neq 0$ so something else must be. For example

$$
\begin{array}{rcll}
\phi_a(x) & = & \phi_a(y) & \text{\# to show } \phi_a \text{ is 1-to-1 show that } \phi_a(x) = \phi_a(y) \Rightarrow x = y \\
& \Rightarrow & ax = ay & \text{\# definition of } \phi_a(x) \text{ (Equation 2)} \\
& \Rightarrow & ax - ay = 0 & \text{\# subtract } ay \text{ from both sides} \\
& \Rightarrow & a(x-y) = 0 & \text{\# } \textit{factor out } a \\
& \Rightarrow & x - y = 0 & \text{\# } a \neq 0 \text{ by assumption so } x - y = 0 \\
& \Rightarrow & x = y & \text{\# so } \phi_a \text{ is one-to-one}
\end{array}
$$

Getting back to showing that ϕ_a is a bijection, we next need to show that ϕ_a is onto. To do this pick an arbitrary $y \in G$ (here G is the range). Then $a^{-1}y \in G$ (here G is the domain) and so $\phi_a(a^{-1}y) = a(a^{-1}y)$. Since multiplication is associative we have $\phi_a(a^{-1}y) = a(a^{-1}y) = (aa^{-1})y = y$. So ϕ_a is onto and hence ϕ_a is a bijection.

2.2 Define $H = \{\phi_a \mid a \in G\}$ and show that H is a subgroup of $Sym(G)$

Now we can define $H = \{\phi_a \mid a \in G\}$. Since each element of H is a bijection from G to G and since $Sym(G)$ is the set of all bijections from G to G we know that $H \subseteq G$. To show that H is a subgroup of $Sym(G)$ we also need to show that H is closed under function composition and inversion.

To show closure under function composition we need to show that $\alpha, \beta \in H \Rightarrow \alpha \circ \beta \in H$. To see this consider $\alpha, \beta \in H$. Then there exists $a \in G$ such that $\alpha = \phi_a$. Similarly there exists $b \in G$ such that $\beta = \phi_b$. So we know that

$$\alpha \circ \beta = \phi_a \circ \phi_b \tag{4}$$

and so for any $x \in G$ we have

[1] Note that having a cancellation law is equivalent to saying there are no *zero divisors*.

[2] A ring with multiplicative inverses is called a division ring (or skew field). Example: the quaternions.

$$
\begin{aligned}
(\alpha \circ \beta)(x) &= (\phi_a \circ \phi_b)(x) && \# \ \alpha \circ \beta = \phi_a \circ \phi_b \text{ (Equation 4)} \\
&= \phi_a(\phi_b(x)) && \# \text{ definition of function composition} \\
&= \phi_a(bx) && \# \ \phi_b(x) = bx \text{ (definition of } \phi_b) \\
&= a(bx) && \# \ \phi_a(x) = ax \text{ (definition of } \phi_a) \\
&= (ab)x && \# \text{ multiplication is associative} \\
&= \phi_{ab}(x) && \# \ \phi_g(x) = gx \text{ where } g = ab
\end{aligned} \tag{5}
$$

So $\alpha \circ \beta = \phi_a \circ \phi_b = \phi_{ab}$. Since $ab \in G$ (G is closed under multiplication) we know that $\phi_{ab} \in H$. Now we have $\phi_{ab} \in H$ and $\alpha \circ \beta = \phi_{ab}$ which together imply that $\alpha \circ \beta \in H$. So H is closed under function composition.

To show that H is closed under inversion we need to show that $\alpha \in H \Rightarrow \alpha^{-1} \in H$. To see this consider $\alpha \in H$. Then there exists an $a \in G$ such that $\alpha = \phi_a$. Since $a \in G$ and since G is a group, $a^{-1} \in G$ and so $\phi_{a^{-1}} \in H$. Note further that for any $x \in G$

$$
\begin{aligned}
(\phi_{a^{-1}} \circ \phi_a)(x) &= \phi_{a^{-1}}(\phi_a(x)) && \# \text{ definition: } (f \circ g)(x) = f(g(x)) \\
&= \phi_{a^{-1}}(ax) && \# \ \phi_a(x) = ax \\
&= a^{-1}(ax) && \# \text{ definition: } \phi_{a^{-1}}(x) = a^{-1}(x) \\
&= (a^{-1}a)x && \# \text{ multiplication is associative} \\
&= x && \# \ a^{-1}a = 1
\end{aligned} \tag{6}
$$

and

$$
\begin{aligned}
(\phi_a \circ \phi_{a^{-1}})(x) &= \phi_a(\phi_{a^{-1}}(x)) && \# \text{ definition: } (f \circ g)(x) = f(g(x)) \\
&= \phi_a(a^{-1}x) && \# \ \phi_{a^{-1}}(x) = a^{-1}x \\
&= a(a^{-1}x) && \# \text{ definition: } \phi_a(x) = ax \\
&= (aa^{-1})x && \# \text{ multiplication is associative} \\
&= x && \# \ aa^{-1} = 1
\end{aligned} \tag{7}
$$

Recall that if a function f is a bijection we know $(f^{-1} \circ f)(x) = (f \circ f^{-1})(x) = x$. From (6) and (7) we see that $\phi_{a^{-1}}$ is the inverse of ϕ_a. More specifically $\phi_{a^{-1}} = \phi_a^{-1}$. Since $\alpha = \phi_a$, $\alpha^{-1} = \phi_a^{-1} = \phi_{a^{-1}} \in H$. So H is closed under inversion.

2.3 Define $f : G \to H$ and show that f is a homomorphic bijection

We still need to show a homomorphic bijection f from G to H. One way to do this is to define $f(g) = \phi_g$ for all $g \in G$. Then to show that f is a bijection we need to show that f is both one-to-one and onto.

To see that f is one-to-one consider

$$\begin{array}{rcll} f(a) & = & f(b) & \text{\# to show } f \text{ is 1-to-1 show that } f(a) = f(b) \Rightarrow a = b \\ & \Rightarrow & \phi_a(x) = \phi_b(x) & \text{\# definition of } f(g) : f(g) = \phi_g \text{ for all } g \in G \\ & \Rightarrow & \phi_a(a) = \phi_b(a) & \text{\# evaluate at } a \in G \\ & \Rightarrow & aa = ba & \text{\# definition of } \phi_g : \phi_g(x) = gx \text{ for all } g \in G \\ & \Rightarrow & (aa)a^{-1} = (ba)a^{-1} & \text{\# multiply by } a^{-1}; a \in G \text{ and } G \text{ a group} \Rightarrow a^{-1} \in G \\ & \Rightarrow & a(aa^{-1}) = b(aa^{-1}) & \text{\# multiplication is associative} \\ & \Rightarrow & a = b & \text{\# } aa^{-1} = 1 \end{array}$$

So f is one-to-one.

To show that f is onto, choose a $\alpha \in H$. Then there exists an $a \in G$ such that $\alpha = \phi_a$. However we know that $f(a) = \phi_a$ and $\phi_a = \alpha$ so we know that $f(a) = \alpha$. So f in onto and since we saw that f is one-to-one, f is a bijection.

Finally, to show that f is also a homomorphism we want to show that $f(ab) = f(a) \circ f(b)$. To see this consider that for any $a, b \in G$ we have

$$\begin{array}{rcll} f(ab) & = & \phi_{ab} & \text{\# definition of } f \\ & = & \phi_a \circ \phi_b & \text{\# Equation 5} \\ & = & f(a) \circ f(b) & \text{\# definition of } f \end{array}$$

So f is a homomorphism.

This completes the proof of Cayley's Theorem.

3 Examples

3.1 $(\mathbb{Z}_4, +) \to (G, \cdot)$

Let $(\mathbb{Z}_4, +)$ be the set $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ with addition modulo 4 and let $(G, \cdot)$ be the set $G = \{1, -1, i, -i\}$ (the fourth roots of unity) with the usual multiplication on $\mathbb{C}$. Then $(\mathbb{Z}_4, +) \simeq (G, \cdot)$. To see that $\mathbb{Z}_4$ is isomorphic to G, let $f : \mathbb{Z}_4 \to G$ be the bijection

$$\begin{array}{ccr} 0 & \longrightarrow & 1 \\ 1 & \longrightarrow & i \\ 2 & \longrightarrow & -1 \\ 3 & \longrightarrow & -i \end{array}$$

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Table 1: $\mathbb{Z}_4$

$\cdot$	1	i	-1	$-i$
1	1	i	-1	$-i$
i	i	-1	$-i$	1
-1	-1	$-i$	1	i
$-i$	$-i$	1	i	-1

Table 2: G

Here are the Cayley tables for $\mathbb{Z}_4$ and G:

To show that f is an isomorphism we need to show that f is a homomorphism, that is, that $f(x+y) = f(x) \cdot f(y)$. Since there are only $n^2 = 4^2 = 16$ values for $f(x+y)$ we can just enumerate them:

$$\begin{array}{llllll}
f(0+0) & = f(0) & = 1 & = 1 \cdot 1 & = f(0) \cdot f(0) \\
f(0+1) & = f(1) & = i & = 1 \cdot i & = f(0) \cdot f(1) \\
f(0+2) & = f(2) & = -1 & = 1 \cdot -1 & = f(0) \cdot f(2) \\
f(0+3) & = f(3) & = -i & = 1 \cdot -i & = f(0) \cdot f(3) \\
f(1+0) & = f(1) & = i & = i \cdot 1 & = f(1) \cdot f(0) \\
f(1+1) & = f(2) & = -1 & = i \cdot i & = f(1) \cdot f(1) \\
f(1+2) & = f(3) & = -i & = i \cdot -1 & = f(1) \cdot f(2) \\
f(1+3) & = f(0) & = 1 & = i \cdot -i & = f(1) \cdot f(3) \\
f(2+0) & = f(2) & = -1 & = -1 \cdot 1 & = f(2) \cdot f(0) \\
 & & \vdots & & \\
f(3+3) & = f(2) & = -1 & = -i \cdot -i & = f(3) \cdot f(3)
\end{array}$$

So the bijection $f : \mathbb{Z}_4 \to G$ above is a homomorphism and hence f is a group isomorphism.

3.2 The Klein 4-group

The Klein 4-group is the group $K = \{e, a, b, c\}$ where e is the identity element and the group operation is defined by the Cayley table below (Table 3).

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table 3: The Klein 4-group Operation

Here K is *not* isomorphic to $\mathbb{Z}_4$. To see this notice that there are 24 bijections from $\mathbb{Z}_4$ and K: $|K| = |\mathbb{Z}_4| = 4$ so there are $n! = 4! = 24$ possible bijections from $\mathbb{Z}_4$ to K. Since we need $f(0) = e$ that leaves $3! = 6$ bijections that could be homomorphisms. For example, consider the bijection

$$\begin{array}{ccc} 0 & \longrightarrow & e \\ 1 & \longrightarrow & a \\ 2 & \longrightarrow & c \\ 3 & \longrightarrow & b \end{array}$$

This bijection is not a homomorphism since $f(1+3) = f(4) = f(0) = e$ while $f(1) \cdot f(3) = ab = c$, so $f(1+3) \neq f(1) \cdot f(3)$.

One way to see that $\mathbb{Z}_4$ is not isomorphic to K is to recognize that every element of K satisfies the equation $x \cdot x = e$ (a key property of the Klein 4-group). However not every element of $\mathbb{Z}_4$ satisfies the equation $x + x = 0$.

This gives a clue as to how to prove, by contradiction, that $\mathbb{Z}_4$ is not isomorphic to K. Specifically, suppose that $\mathbb{Z}_4$ is isomorphic to K. Then there exists a bijection $f : \mathbb{Z}_4 \to K$ such that $f(x+y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{Z}_4$. Well, we know by definition that $f(0) = e$ and since f is one-to-one we also know that $f(1) \neq e$. Since f is a homomorphism we also know that

$$f(1+1) = f(1) \cdot f(1)$$

However, since $f(1) \in K$ and all elements of K satisfy $x \cdot x = e$ we can conclude that $f(1) \cdot f(1) = e$, so $f(1+1) = f(2) = e$. Now we have $f(0) = e$ *and* $f(2) = e$ which is a contradiction since we assumed that f was one-to-one. So the original assumption that $\mathbb{Z}_4$ is isomorphic to K is false.

Ok, but Cayley's Theorem says there is a subgroup H of S_4 which is isomorphic to K. How to find H? Since as noted above Cayley's Theorem is constructive, we should be able to follow the approach used in the proof to find H. Here we let $H = \{\phi_e, \phi_a, \phi_b, \phi_c\}$ where, for all $x \in K$

$$\begin{array}{ll} \phi_e(x) = ex & \qquad \# \ \phi_e(x) = x \\ \phi_a(x) = ax & \\ \phi_b(x) = bx & \\ \phi_c(x) = cx & \end{array}$$

Now we can rewrite the Cayley table for the Klein 4-group (Table 3) as

$$\phi_e = \begin{pmatrix} e & a & b & c \\ e & a & b & c \end{pmatrix}$$
$$\phi_a = \begin{pmatrix} e & a & b & c \\ a & e & c & b \end{pmatrix}$$
$$\phi_b = \begin{pmatrix} e & a & b & c \\ b & c & e & a \end{pmatrix}$$
$$\phi_c = \begin{pmatrix} e & a & b & c \\ c & b & a & e \end{pmatrix}$$

Now, if we relabel K by the bijection

$$\begin{array}{cccc} e & a & b & c \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 2 & 3 & 4 \end{array}$$

we can represent K in cyclic notation:

$$\phi_e = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix} = (1)$$
$$\phi_a = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} = (12)(34)$$
$$\phi_b = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} = (13)(24)$$
$$\phi_c = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} = (14)(23)$$

Now we can see that $K \simeq H$ where $H = \{(1), (12)(34), (13)(24), (14)(23)\}$. That is, $f : K \to H$ is the bijection

$$f: \begin{array}{cccc} 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ (1) & (12)(34) & (13)(24) & (14)(23) \end{array}$$

The Klein 4-group K has many other interesting properties, including

- K is the smallest non-cyclic group
- K is the underlying group of the four-element field

- K is the symmetry group of a non-square rectangle
- K is the group of bitwise exclusive or operations on two-bit binary values
- $K = \mathbb{Z}_2 \times \mathbb{Z}_2$, the direct product of two copies of the cyclic group of order 2

4 Acknowledgements

References